

BMS INSTITUTE OF TECHNOLOGY AND MANAGEMENT

(An Autonomous Institute affiliated to VTU, Belagavi, Approved by AICTE New Delhi)

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Department of Computer Science and Engineering



Mini Project Synopsis

Academic Year 2026-27

Batch No: 8	Guide Name: Dr. Ravi Hosur	Submission Date:
Mini Project Title Sign-Language Spelling Assistant: Indian Sign Language Alphabet Recognition and Word Formation		
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Project Category		Application Project

Signature of Guide

Abstract:

Indian Sign Language fingerspelling involves representing words through a sequence of individual alphabet gestures. Recognizing and processing these gestures continuously can be challenging when the input is incomplete or incorrectly recognized. The proposed Sign-Language Spelling Assistant is an AI-based system designed to support users during the fingerspelling process. The system captures hand gestures through a camera, recognizes the corresponding ISL alphabet signs, and processes them sequentially to form words. It further assists with word prediction, completion, error correction, confidence feedback, and letter editing. The recognized words can be combined to form sentences and converted into text and voice output. The system also supports personalized vocabulary and practice features to improve the overall fingerspelling experience.

Introduction:

Indian Sign Language (ISL) is a visual language that enables communication through hand gestures and movements. Fingerspelling is an important method of representing words by performing individual alphabet signs one after another, especially when expressing names, technical terms, uncommon words, or words that may not be available in a predefined vocabulary. The Sign-Language Spelling Assistant focuses on making this fingerspelling process more interactive and useful. The system captures the user's hand gestures through a camera, recognizes the corresponding ISL alphabets, and interprets the sequence as meaningful text. It can assist the user through word prediction, word completion, error correction, confidence feedback, and letter editing. The recognized words can also be combined to form sentences and converted into voice output. In addition, features such as custom vocabulary, word history, and a practice mode provide personalization and learning support. Overall, the project extends basic ISL alphabet recognition into an interactive assistant for fingerspelling, communication, and practice.

Motivation:

Sign-language recognition systems can assist communication by converting gestures into text, but fingerspelling involves a continuous sequence of individual alphabet signs rather than a single predefined gesture. Recognizing and combining these letters correctly can become difficult when the input is continuous, incomplete, or incorrectly recognized. This motivates the development of a spelling assistant that can process alphabet gestures sequentially and provide support during word formation, rather than limiting the user to direct recognition of individual signs.

Therefore, the project aims to make fingerspelling more flexible, interactive, and user-oriented.

Problem Statement:

To develop a computer-vision-based Sign-Language Spelling Assistant that can recognize ISL alphabet gestures during continuous fingerspelling and maintain their correct sequence for word formation, while addressing challenges such as incomplete input, recognition errors, and uncertainty during the spelling process.

Objectives:

1. To overcome the challenge of accurately recognizing individual ISL alphabet gestures from real-time camera input during continuous fingerspelling.
2. To address errors in gesture sequencing and incomplete word recognition while converting continuous fingerspelling into meaningful words and sentences.
3. To overcome ambiguity and recognition errors by identifying low confidence gestures and enabling correction of incorrectly recognized letters.
4. To address the difficulty of adapting ISL recognition to individual users and their frequently used or customized vocabulary.
5. To bridge the gap between recognized ISL content and effective communication and learning through spoken output and real-time performance feedback.

SDG Sustainable Development Goal:

SDG 10 – Reduced Inequalities

The Sign Language Spelling Assistant is mapped to SDG 10 - Reduced Inequalities, particularly SDG 10.2 - Social and Economic Inclusion, as it aims to reduce communication barriers faced by individuals who use Indian Sign Language (ISL). Communication difficulties can limit interaction between ISL users and people who do not understand sign language, affecting participation in education, services, workplaces, and everyday social situations.

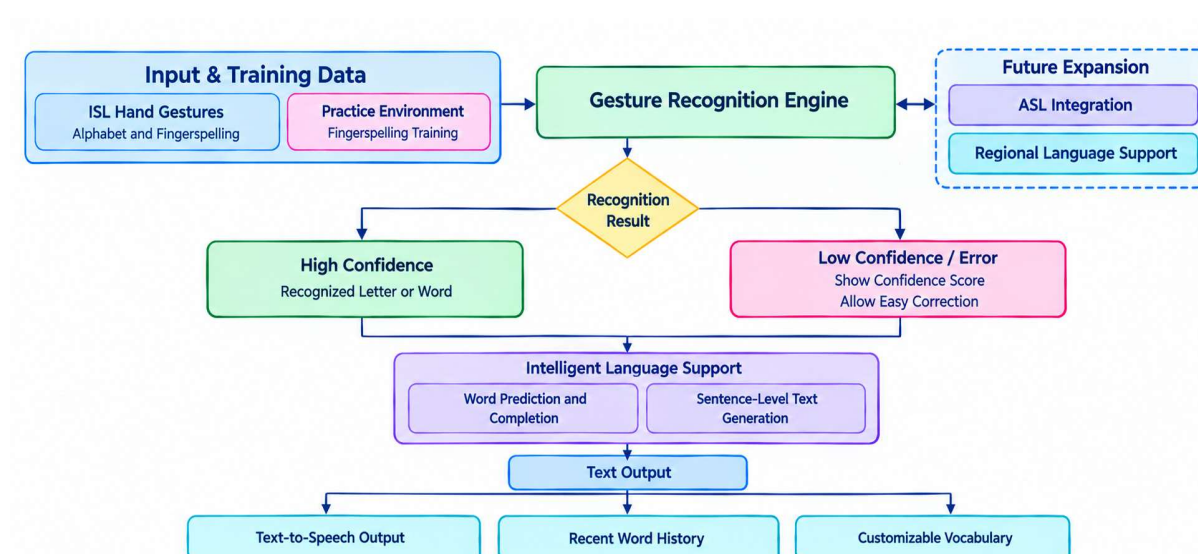
The proposed system uses computer vision and machine learning to recognize ISL alphabet gestures and convert them into text, with additional support for word prediction, sentence formation, correction, and text-to-speech. These capabilities can make ISL-based communication more accessible to a wider range of users.

By providing a technology based communication aid, the project supports inclusive participation, accessibility, and equal opportunities, aligning with the broader objective of SDG 10 to reduce inequalities and promote the social inclusion of people regardless of their circumstances.

Expected Outcome:

The system will provide reliable ISL gesture recognition and continuous fingerspelling, with word prediction, error correction, sentence generation, and text-to-speech. It will also support custom vocabulary, word history, and practice mode.

Future Scope: Integration with ASL and regional languages for broader accessibility.



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